Al- Balqa' Applied University
Faculty of Engineering Technology
Department of Physics and Basic Sciences

جامعة البلقاء التطبيقية
كلية الهندسة التكنولوجية
قسم العلوم الفيزياء والعلوم الأساسية

Physics 111
Final Exam
Time: 90 minutes (B)

موعد المختبر : مدرس المختبر: الاسم بالعربية:

Q1) Circle the correct answer **(32 degrees)**

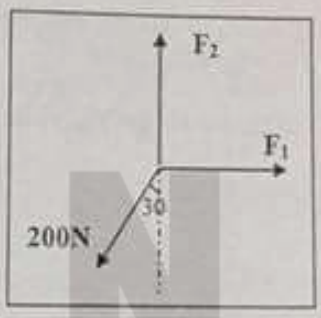
1. Two forces F_1 , F_2 are equal to each other and to their resultant, the angle between the two forces is:

a) 120^0 b) 45^0 c) 180^0 d) 90^0

2. The set of forces shown in the following figure are in equilibrium, the magnitudes of F_1 and F_2 (in N) are respectively given by:

(a) (100,173) (b) (173,100)

(c) (-100,-173) (d) (-173,-100)



3. A Vernier caliber was used to measure the length of a cube, the error in length (ΔL) is given by:

a) 0.1mm b) 0.05mm c) 0.5mm d) 0.005mm

4. Which of the following measurements represents the reading of the micrometer?

a) 7.28 mm b) 8.0 cm c) 1.50 cm d) 2 mm

5. In spring constant experiment, if the hangings mass 80g and the elongation in the spring is 2cm, the spring constant (N/m) is:

a) 5 b) 40 c) 20 d) 5

6. A simple pendulum has a simple harmonic motion if:

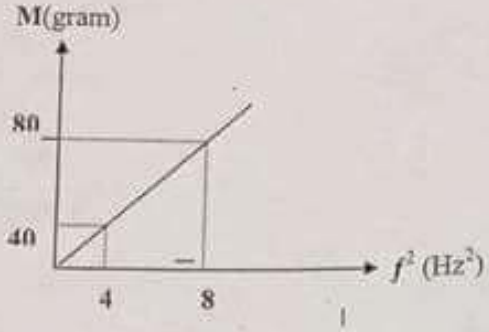
a) The string is very long.
b) The angle gives $\sin\theta \approx \theta$.
c) The mass to length ratio is small.
d) The mass to length ratio is large.

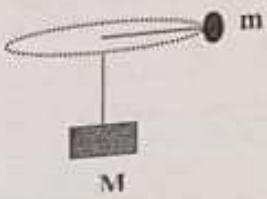
7. In friction experiment, if the coefficient of kinetic friction is $\mu_k = 0.2$, the force required to keep a block of mass 0.3 kg moving horizontally with constant speed is equal to:

a) 2.94 N b) 0.60 N c) 14.7 N d) 0.90 N

GNE COMMUNITY NETWORK ENGINEERING

8. In the circular motion experiment, a student studied the relation between the hanging mass (M) and the square of the frequency (f^2) for rotational body (m) using the radius (R=0.6 m). A plotted data is shown in the figure bellow.

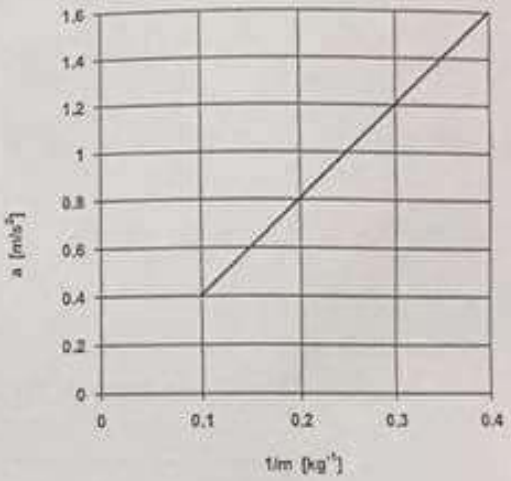




The mass (m) of the rotational body is:

a) 6.34 gram b) 12.7 gram c) 8.44 gram d) 4.22 gram

In the Newton's second law experiment, the relation between the acceleration (a) and the total mass (m) with a constant force (F) is shown in Figure.



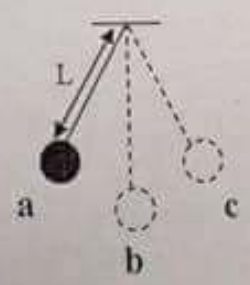
9. The applied force on the system (in N) is:

a) 4 b) 12
c) 16 d) 8

10. The hanging mass m_h (in gram) is:

a) 800 b) 120
c) 400 d) 160

In the figure shown, the bob takes 0.749 seconds to move from point a to b. If the length of the string is 2 m, then



11. The acceleration due to gravity, g, (in m/s^2) is:

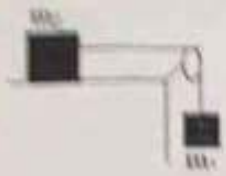
a) 9.2 b) 8.8

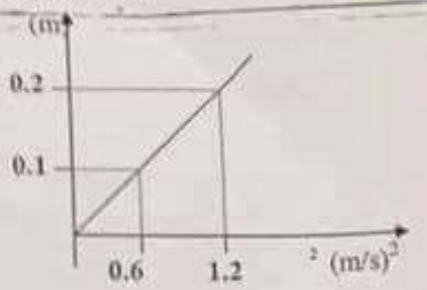
c) 10 d) 9.6

12. If we doubled the mass of bob, then the period (T) will be:

a) 2T b) T c) T/2 d) $(T)^{1/2}$

14. In the conservation of energy experiment, a student obtained data plotted in the following figure. If the masses m_1=0.15 kg and m_2=0.2 kg, the acceleration of gravity is:





a) 7 m/s² b) 11 m/s² c) 9 m/s² d) 13m/s²

15. In the collision between two carts, all of the velocity was transferred from the first cart to the second cart. What type of collision is this?

a) Perfectly inelastic collision b) Inelastic collision c) Elastic collision d) momentum is not conserved

16. In the collision between two carts, the carts combined after the collision with one third of the velocity of the first cart. How much kinetic energy is lost if each cart has a mass m and the initial velocity of the first cart was v? (v_2 = 0)

a) $\frac{1}{3}mv^2$ b) 0 c) $\frac{8}{9}mv^2$ d) $\frac{7}{18}mv^2$

18. In uniform circular motion, if the radius of circular path is 20 cm and time needed for 50 periods was 25 sec, the Speed of the object and its centripetal acceleration (in m/s).

a) 0.63 b) 1.26 c) 2.51 d) 5.03

COMPUTER NETWORK ENGINEERING

Q2: Simple Harmonic Motion Experiment

a) Calculate the spring constant from the following Data:

Load mass m (kg)	Period of oscillation T (s)	Spring constant k (N/m)
0.1	0.27	
0.2	0.40	
0.3	0.51	

b) Calculate the average value of the spring constant?

$\bar{k}$ = _____

6) Using significant rules to find 30600/45.0 = (680).

7) In gravitational potential energy experiment, before start move toward down from (h_i=60cm) the mass was (m=20g) and (M=200g) ,then the kinetic energy when (m) reach (h=40cm) is: (0.04 joule).

8) For the above question the final velocity (V_f) when m reach ground is: (1.04 m/s).

9) Using significant rules to find , 3.924+3.78= (7.70).

- *Q2 : Circle the correct answer:(10.5 marks)*

1) If a student used a ruler with a maximum range (10 cm) to measure the length of a table which has the length (1.65 m), then the uncertainty in length (ΔL) is:

a) 3mm b) 4.5mm (c) 8.5mm d) 6mm

2) If the relation between A&B given as (C =A-B), if A = 5.28 and B=1.8,then (C±ΔC) is:

(Using significant rules)

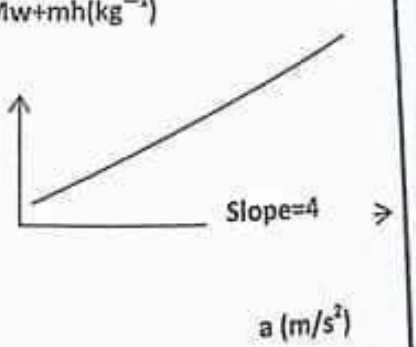
a) 3.48±0.05 b) 3.5±0.1 (c) 3.5±0.06 d) 3.48±0.055

3) F1,F2,F3 are three forces in equilibrium ,if you know that (F_{3x} =1.32 N , F_{3y} =-7.5 N),the direction of resultant between (F1,& F2) according to x-positive axis is :

(a) 100 b) 80 c) 280 d) 260

4) In Newton's second law experiment, if we got this graph,

then the constant hanged mass in the experiment is:



a)15g (b)25g

c)60g d)80g

15. In friction experiment, if a horizontal force of 0.6 N is required to set the block of mass 0.2 kg in motion, the coefficients of static friction μ_s equal to:

a) 0.6 b) 0.2 c) 0.3 d) 0.4

16. Two identical blocks (with the same mass) are pulled along a rough surface as suggested in the figure.



Which one of the following statements is false?
(a) The coefficient of kinetic friction is the same in each case.
(b) A force of the same magnitude is needed to keep each block moving.
(c) A force of the same magnitude was required to start each block moving.
(d) The normal force exerted on the blocks by the surface is the same for both blocks.
(e) The magnitude of the force of kinetic friction is greater for the block on the right.

17. In friction experiment, if the coefficient of kinetic friction $\mu_k = 0.2$ of a horizontal surface, the required to keep a block of mass 0.3 kg moving with constant speed is equal:

a) 2.94 N b) 0.60 N c) 14.7 N d) 0.90 N

18. If F is the force, x the displacement, and k a spring constant, for simple harmonic motion F is given

a) $F = kx$ b) $F = -\frac{k}{x}$ c) $F = -kx^2$ d) $F = -kx$

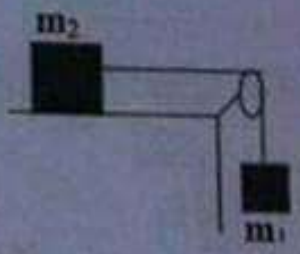
19. In elasticity experiment, if the hanging mass 80g and the elongation in the spring is 4cm, the spring constant (N/m) is:

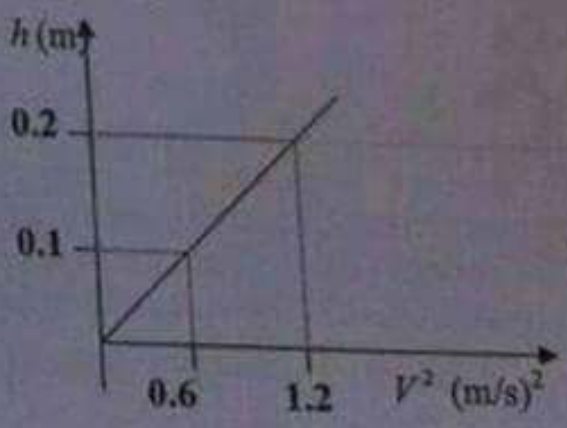
a) 5 b) 40 c) 20 d) 5

20. In elasticity experiment, if spring constant k = 20 N/m and the time of ten oscillations is 4.44sec, of hanging weight (in gram) is:

a) 100 b) 195 c) 225 d) 75

21. In the conservation of energy experiment, a student obtained data plotted in the following figure. masses m_1=0.15 kg and m_2=0.2 kg. The acceleration of gravity is:





a) 7 m/s² b) 11 m/s² c) 9 m/s² d) 13m/s²

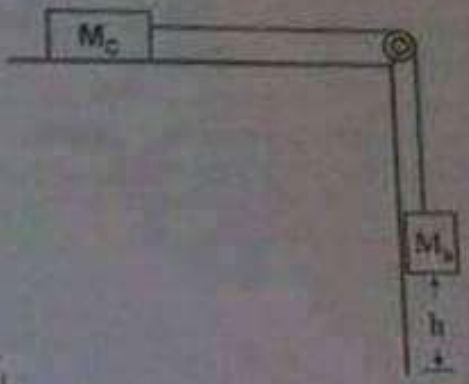
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Physics 111
Final Exam
Time: 90 minutes (C)

الاسم بالعربية: مدرس المختبر: موعد المختبر :

A) A hanging mass $M_h = 0.5kg$, and the mass of the cart $M_c = 1kg$. The system is released from rest. The height of the hanging mass is **h=50 cm**.



a) Complete the following table:

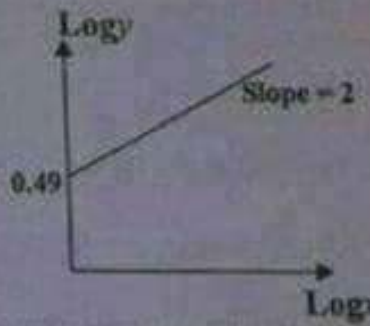
Where K is the kinetic of the system, U is the potential Energy.

U_i (J)	U_f (J)	K_i (J)	K_f (J)

The speed of the cart v=.....m/s, as the hanging mass just hits the ground.

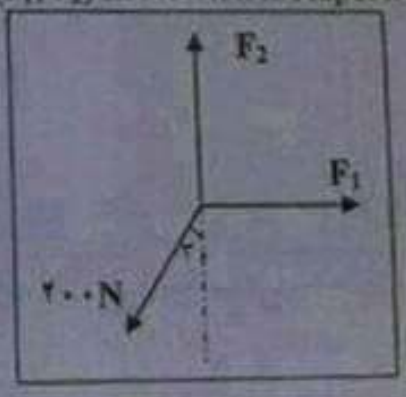
B) Circle the correct answer

1. The figure shows the relation between log x and Log y, then:



a) $Y = 2X^3 + 2$ b) $Y = 3X^2$
c) $Y = 10X^3$ d) $Y = 3X^{10}$

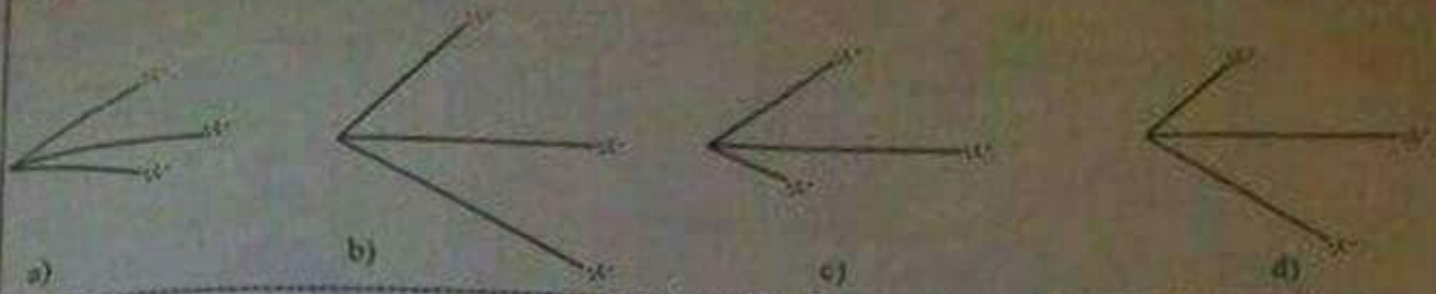
2. The set of forces shown are in equilibrium, the Values of (F_1, F_2) in Newton is respectively given by:



(a) (100,173) (b) (173,100)

(c) (-100,-173) (d) (-173,-100)

10. In an elastic collision of two steel balls of equal masses, which of the following represents the correct experimental situation?

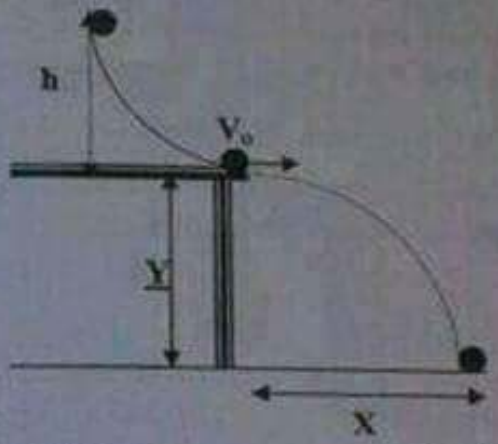


a) b) c) d)

11. One of the following statements is **true**

a) In the elastic collisions, the linear momentum is conserved, but the kinetic energy is not.
b) In inelastic collisions, the linear momentum and the kinetic energy are not conserved.
c) The linear momentum is conserved for both elastic and inelastic collisions.
d) The kinetic energy is conserved for both elastic and inelastic collisions.

12. In the collisions in two dimensions' experiment, a student obtained the results as shown in the figure for the mass M_1=10g when dropped from a height h to calculate its velocity before collisions with another mass, for Y= 80cm, and X= 40 cm.

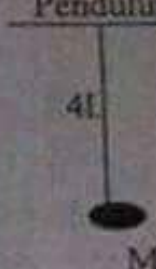


The velocity (Vo) is:
a) 2.0 m/s b) 1.0 m/s c) 4.0 m/s d) 1.4 m/s

The height h equals _____

13. The motion of a simple pendulum is approximately simple harmonic if and only if:
a) The string is very long.
b) The angle gives $\sin\theta \approx \theta$.
c) The mass to length ratio is small.
d) The mass to length ratio is large.

14. In the simple pendulum experiment, the following figures show two pendulums. If the periodic time is T_1 for the Pendulum (1), and the periodic time is T_2 for the pendulum (2). Then the relation between T_1 and T_2 is



Pendul
L

a) $T_2 = 2T_1$ b) $T_2 = 1.73T_1$ c) $T_2 = 2.24T_1$ d) $T_2 = 2.45T_1$

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Physics 111
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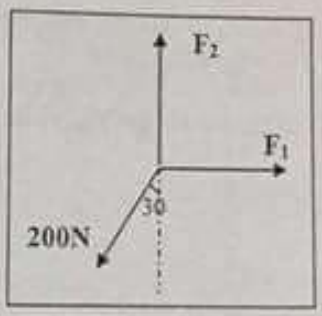
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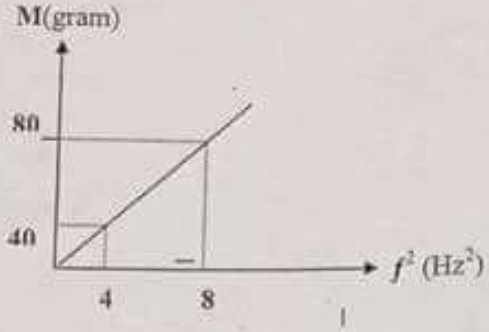
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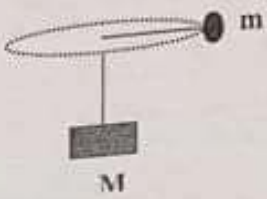
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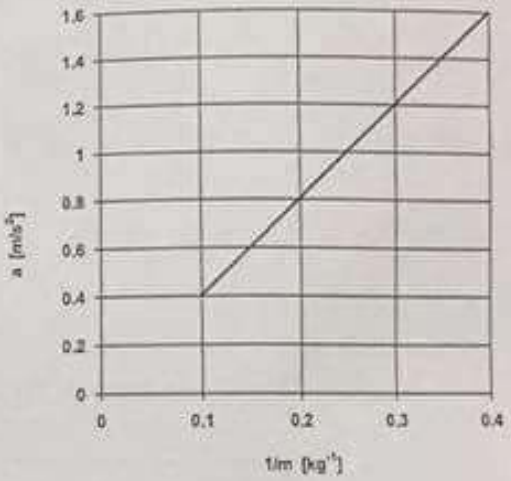




The mass (m) of the rotational body is:

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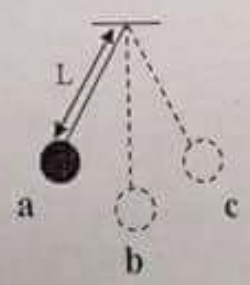
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a) 800 b) 120
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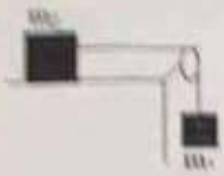
a) 9.2 b) 8.8

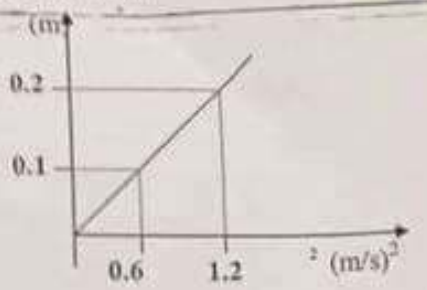
c) 10 d) 9.6

12. If we doubled the mass of bob, then the period (T) will be:

a) 2T b) T c) T/2 d) $(T)^{1/2}$

14. In the conservation of energy experiment, a student obtained data plotted in the following figure. If the masses m_1=0.15 kg and m_2=0.2 kg, the acceleration of gravity is:





a) 7 m/s² b) 11 m/s² c) 9 m/s² d) 13m/s²

15. In the collision between two carts, all of the velocity was transferred from the first cart to the second cart. What type of collision is this?

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16. In the collision between two carts, the carts combined after the collision with one third of the velocity of the first cart. How much kinetic energy is lost if each cart has a mass m and the initial velocity of the first cart was v? ($v_2 = 0$)

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18. In uniform circular motion, if the radius of circular path is 20 cm and time needed for 50 periods was 25 sec, the Speed of the object and its centripetal acceleration (in m/s).

a) 0.63 b) 1.26 c) 2.51 d) 5.03

Q2: Simple Harmonic Motion Experiment

a) Calculate the spring constant from the following Data:

Load mass m (kg)	Period of oscillation T (s)	Spring constant k (N/m)
0.1	0.27	
0.2	0.40	
0.3	0.51	

b) Calculate the average value of the spring constant?

$\bar{k}$ = _____

B

Q3: You are going to give the cart 1 a push and collide it with cart 2, which is initially at rest. After the collision cart 1 goes in the opposite direction. Use the following data to complete the table for cart 1 and cart 2 before and after the collision:

The mass $m_1 = 100$ g

The mass $m_2 = 200$ g

The Distance L = 4 cm

t_{1i}	t_{1r}	t_{2r}
0.165 s	0.485 s	0.246 s

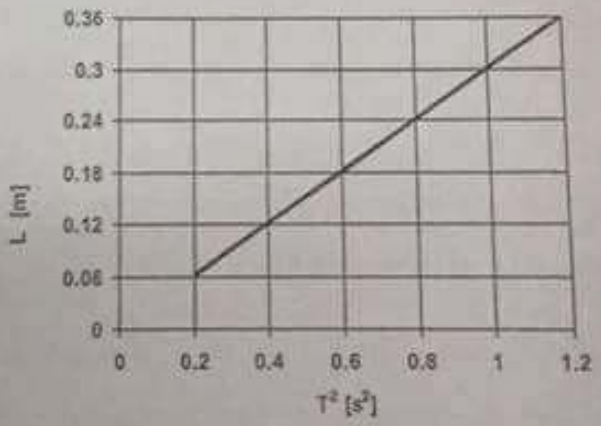
a) Complete the following table:

	Before Collision	After Collision
The velocity of cart 1 v_1 [m/s]		
The velocity of cart 2 v_2 [m/s]		
The total momentum P [kg.m/s]		
The total kinetic K.E [J]		

b) Is the total momentum conserved? ----------

c) Is the collision elastic or inelastic? ----------

Q4: In simple pendulum experiment, a student plots a relation between the length of the pendulum and the square of the period (T^2) as in figure shown. Answer the following questions.



a) Write the relation between the slope of the figure and the acceleration of gravity? ----------

b) Determine the experimental value for g? ----------

c) If the accepted value for g in this lab is 9.8 m/sec^2, what is the percent error for this experiment?

) The period (T) when the length is 18 cm. ----------

6) Using significant rules to find 30600/45.0 = (680).

7) In gravitational potential energy experiment, before start move toward down from (h_i=60cm) the mass was (m=20g) and (M=200g) ,then the kinetic energy when (m) reach (h=40cm) is: (0.04 joule).

8) For the above question the final velocity (V_f) when m reach ground is: (1.04 m/s).

9) Using significant rules to find , 3.924+3.78= (7.70).

- *Q2 : Circle the correct answer:(10.5 marks)*

1) If a student used a ruler with a maximum range (10 cm) to measure the length of a table which has the length (1.65 m), then the uncertainty in length (ΔL) is:

a) 3mm b) 4.5mm (c) 8.5mm d) 6mm

2) If the relation between A&B given as (C =A-B), if A = 5.28 and B=1.8,then (C±ΔC) is:

(Using significant rules)

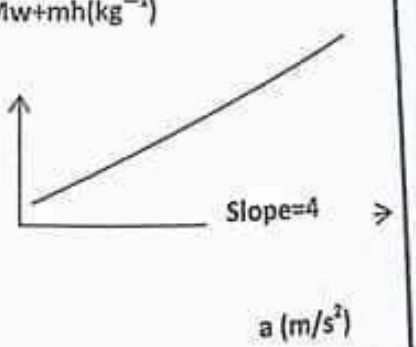
a) 3.48±0.05 b) 3.5±0.1 (c) 3.5±0.06 d) 3.48±0.055

3) F1,F2,F3 are three forces in equilibrium ,if you know that (F_{3x} =1.32 N , F_{3y} =-7.5 N),the direction of resultant between (F1,& F2) according to x-positive axis is :

(a) 100 b) 80 c) 280 d) 260

4) In Newton's second law experiment, if we got this graph,

then the constant hanged mass in the experiment is:



a)15g (b)25g

c)60g d)80g

15. In friction experiment, if a horizontal force of 0.6 N is required to set the block of mass 0.2 kg in motion, the coefficients of static friction μ_s equal to:

a) 0.6 b) 0.2 c) 0.3 d) 0.4

16. Two identical blocks (with the same mass) are pulled along a rough surface as suggested in the figure.



Which one of the following statements is false?
(a) The coefficient of kinetic friction is the same in each case.
(b) A force of the same magnitude is needed to keep each block moving.
(c) A force of the same magnitude was required to start each block moving.
(d) The normal force exerted on the blocks by the surface is the same for both blocks.
(e) The magnitude of the force of kinetic friction is greater for the block on the right.

17. In friction experiment, if the coefficient of kinetic friction $\mu_k = 0.2$ of a horizontal surface, the required to keep a block of mass 0.3 kg moving with constant speed is equal:

a) 2.94 N b) 0.60 N c) 14.7 N d) 0.90 N

18. If F is the force, x the displacement, and k a spring constant, for simple harmonic motion F is given

a) $F = kx$ b) $F = -\frac{k}{x}$ c) $F = -kx^2$ d) $F = -kx$

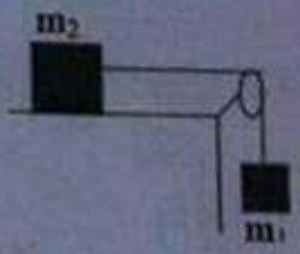
19. In elasticity experiment, if the hanging mass 80g and the elongation in the spring is 4cm, the spring constant (N/m) is:

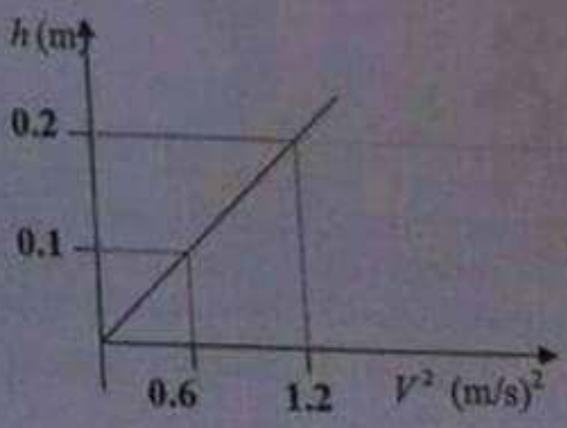
a) 5 b) 40 c) 20 d) 5

20. In elasticity experiment, if spring constant k = 20 N/m and the time of ten oscillations is 4.44sec, of hanging weight (in gram) is:

a) 100 b) 195 c) 225 d) 75

21. In the conservation of energy experiment, a student obtained data plotted in the following figure. masses m_1=0.15 kg and m_2=0.2 kg. The acceleration of gravity is:





a) 7 m/s² b) 11 m/s² c) 9 m/s² d) 13m/s²

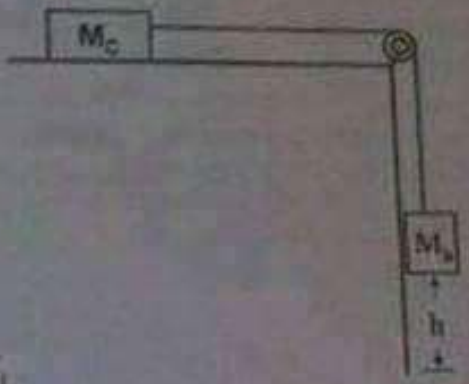
Al- Balqa' Applied University
Faculty of Engineering Technology
Department of Applied Science

جامعة البلقاء التطبيقية
كلية الهندسة التكنولوجية
قسم العلوم التطبيقية

Physics 111
Final Exam
Time: 90 minutes (C)

الاسم بالعربية: مدرس المختبر: موعد المختبر :

A) A hanging mass $M_h = 0.5kg$, and the mass of the cart $M_c = 1kg$. The system is released from rest. The height of the hanging mass is **h=50 cm**.



a) Complete the following table:

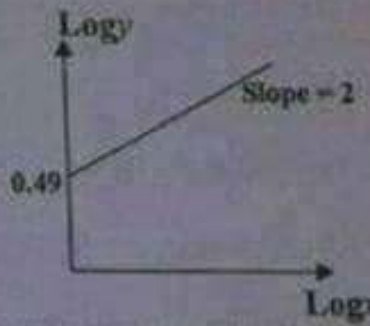
Where K is the kinetic of the system, U is the potential Energy.

U_i (J)	U_f (J)	K_i (J)	K_f (J)

The speed of the cart v=.....m/s, as the hanging mass just hits the ground.

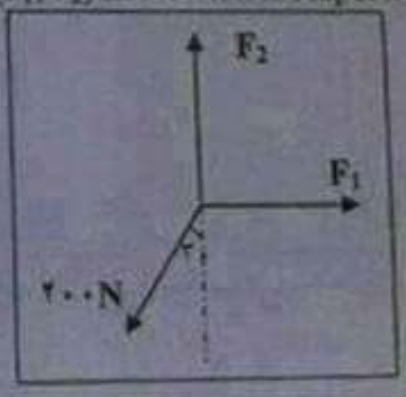
B) Circle the correct answer

1. The figure shows the relation between log x and Log y, then:



a) $Y = 2X^3 + 2$ b) $Y = 3X^2$
c) $Y = 10X^3$ d) $Y = 3X^{10}$

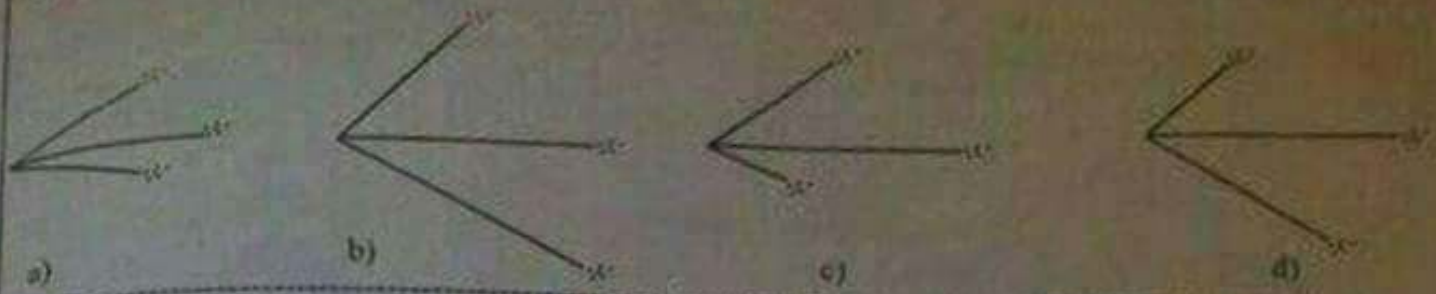
2. The set of forces shown are in equilibrium, the Values of (F_1, F_2) in Newton is respectively given by:



(a) (100,173) (b) (173,100)

(c) (-100,-173) (d) (-173,-100)

10. In an elastic collision of two steel balls of equal masses, which of the following represents the correct experimental situation?

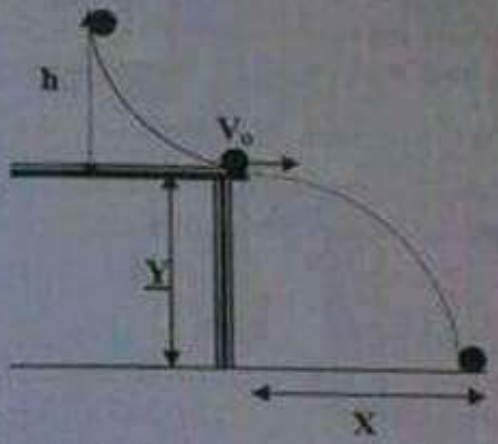


a) b) c) d)

11. One of the following statements is **true**

a) In the elastic collisions, the linear momentum is conserved, but the kinetic energy is not.
b) In inelastic collisions, the linear momentum and the kinetic energy are not conserved.
c) The linear momentum is conserved for both elastic and inelastic collisions.
d) The kinetic energy is conserved for both elastic and inelastic collisions.

12. In the collisions in two dimensions' experiment, a student obtained the results as shown in the figure for the mass M_1=10g when dropped from a height h to calculate its velocity before collisions with another mass, for Y= 80cm, and X= 40 cm.

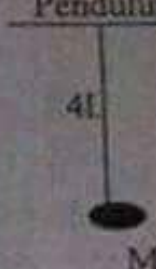


The velocity (Vo) is:
a) 2.0 m/s b) 1.0 m/s c) 4.0 m/s d) 1.4 m/s

The height h equals _____

13. The motion of a simple pendulum is approximately simple harmonic if and only if:
a) The string is very long.
b) The angle gives $\sin\theta \approx \theta$.
c) The mass to length ratio is small.
d) The mass to length ratio is large.

14. In the simple pendulum experiment, the following figures show two pendulums. If the periodic time is T_1 for the Pendulum (1), and the periodic time is T_2 for the pendulum (2). Then the relation between T_1 and T_2 is



Pendul
L

a) $T_2 = 2T_1$ b) $T_2 = 1.73T_1$ c) $T_2 = 2.24T_1$ d) $T_2 = 2.45T_1$

3. A Vernier caliper was used to measure the length of a cube, the error in length (ΔL) is:

a) 0.1mm b) 0.05mm c) 0.5mm d) 0.005mm

4. Which of the following measurements represents the reading of the micrometer?

a) 9.25 mm b) 8.0 cm c) 1.50 cm d) 2 mm

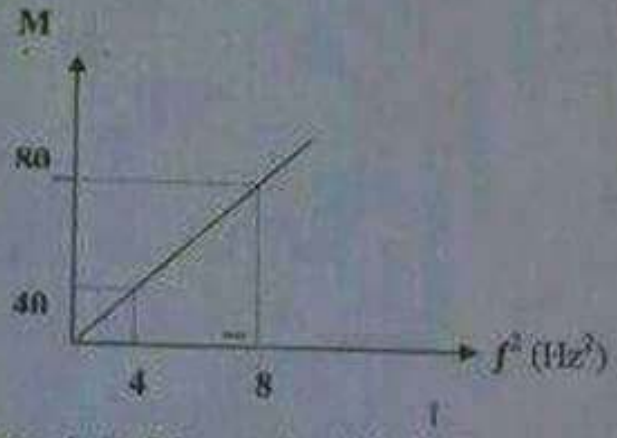
5. A student wanted to calculate the density (ρ) of Mass (M). If $M \pm \Delta M = (10 \pm 0.01)$ gram and its volume $V \pm \Delta V = (5 \pm 0.1)\ cm^3$, the absolute error of the density ($\Delta\rho$) in g/cm^3 is:

a) 0.08 b) 0.04 c) 0.06 d) 0.02

6. An object is moving with a constant speed in a circular path of radius r. If its frequency of revolution is (2n), express its centripetal acceleration in terms of n and r.

a) $4\pi^2n^2r$ b) $16\pi^2n^2r$ c) $8\pi^2n^2r$ d) $2\pi^2n^2r$

7. In the circular motion experiment, a student studied the relation between the hanging mass (M) and the square of the frequency (f^2) for rotational body (m) by making the radius (R=0.6 m). A plotted data is shown in the figure bellow.

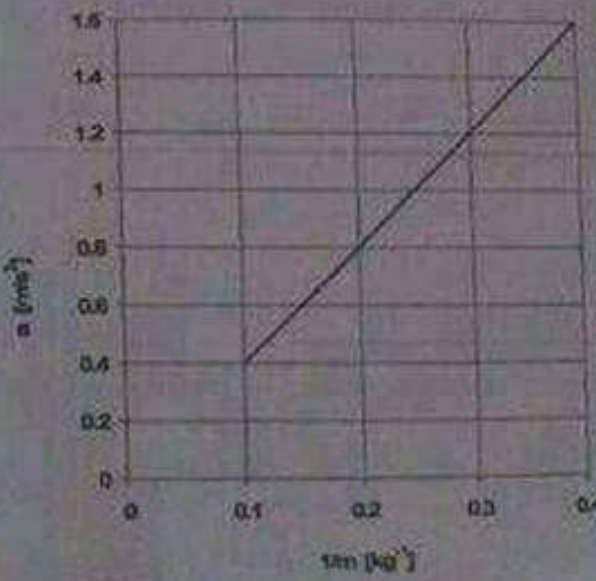




The mass (m) of the rotational body is:

a) 6.34 gram b) 12.7 gram c) 8.44 gram d) 4.22 gram

In the Newton's second law experiment, the relation between the acceleration (a) and the total mass (m) with a constant force (F) is shown in Figure.



8. The applied force on the system (in N) is:

a) 4 b) 12
c) 16 d) 8

9. The hanging mass m_s (in gram) is:

a) 800 b) 120
c) 400 d) 160

4) a) $9.25\,mm \rightarrow 0.925\,cm$

5) $M \pm \Delta M \quad (10 \pm 0.01)\,g$

$V \pm \Delta V \quad (5 \pm 0.1)\,cm^3$

$\Delta\rho$

$$\rho = \frac{m}{V} \Rightarrow \frac{10}{5} = 2\,g/cm^3$$

$$\frac{\Delta\rho}{\rho} = \frac{\Delta m}{m} + \frac{\Delta V}{V}$$

$$\Delta\rho \Rightarrow 2\left(\frac{0.01}{10} + \frac{0.1}{5}\right)$$

$$\Rightarrow 0.04 \quad (b)$$

6) $f = 2n$

$$a_c \Rightarrow \frac{v^2}{r} \Rightarrow \frac{(2\pi r f)^2}{r}$$

$$4\pi^2 r f^2 \Rightarrow 4\pi^2 r \cdot 4n^2$$

$$16\pi^2 r n^2$$

(b)

21)

$m_1 = 0.15\ kg$

$m_2 = 0.2\ kg$

$K = U$

$\frac{1}{2}(m_1+m_2)v^2 = m_1 g h$

$\frac{\Delta h}{\Delta v^2} \Rightarrow \frac{\frac{1}{2}(m_1+m_2)}{m_1 g}$

slope

$\frac{0.2 - 0.1}{1.2 - 0.6} = 0.1666$

$0.16666 \Rightarrow \frac{1}{2}\frac{(0.15+0.2)}{0.15\ g}$

~~$g = \frac{1.666 \times 0.15}{0.5 \times 0.35}$~~

$0.166 * 0.15\ g = \frac{1}{2}(0.35)$

$g \Rightarrow 7\ m/s$

A) #

18) d) $F = -kx$

19)

$m_{ASI} = 80g \longrightarrow 0.08\ kg$

$\Delta x = 4\ cm \longrightarrow 0.04\ m$

$F = k\,\Delta x$

mg

$\frac{0.8}{0.04} = 20$

C #

20) $k = 20\ N/m$

$T = \frac{4.44}{10} = 0.444\ sec$

$\omega^2 = \frac{k}{m}$

$\left(\frac{2\pi}{T}\right)^2 = \frac{k}{m} \Rightarrow \frac{4\pi^2}{T^2} = \frac{k}{m}$

$m = \frac{T^2 k}{4\pi^2} \Rightarrow \approx 0.998\ kg$

$99.8\ g \approx 100$

A #

11) (b)

13) (b)

14) (A) $T_2 = 2T_1$

15) $F = 0.6\ N \qquad m = 0.2\ kg$

$\mu_s =$

$F_s = \mu_s N$

$0.6 = \mu_s (0.2 * 10)$

$\mu_s = 0.3$ C #

16) (d)

17) $\mu_k = 0.2 \qquad m = 0.3$

$F_k = \mu_k N$

$(0.2)(0.3)(10)$

$= 0.60\ N$ b #

2)

$\Sigma f_x = \text{zero}$

$-200 \sin 30 + F_1 = \text{zero}$

$F_1 = 100\ N$

$\Sigma f_y = \text{zero}$

$-200 \cos 30 + F_2 = \text{zero}$

$F_2 = 173\ N$

الجواب: (a) (100, 173)

3)

(b) 0.05 mm

~~(c) 150 cm = 0.15~~

~~9.25 → 9~~

A) $M_h = 0.5kg$ $M_c = 1kg$ $h = 50cm = 0.5m$

$K_i = zero$

$U_f = zero$

$U_i = M_h g h \Rightarrow (0.5)(9.8)(0.5)$

$\Rightarrow 2.45\ J$

$\Delta K + \Delta U = 0$

$(K_f - K_i) + (U_f - U_i) = 0$

$K_f - U_i = 0$

$\frac{1}{2}(M_h + M_c)V_f^2 = M_h g h$

$\frac{1}{2}(1+0.5)V_f^2 = 2.45$

$V_f \Rightarrow 1.80\ m/s$

$K_f \Rightarrow \frac{1}{2}(1+0.5)(1.8)^2 \Rightarrow 2.45$

$y = 3x^2 \rightarrow$ a)

$\log y = \log 3 + \log x^2$

$\log y = (0.49) + (2)\log x$

الجواب (b) #

الإزاحة — slope

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Faculty of Engineering Technology
Department of Applied Science

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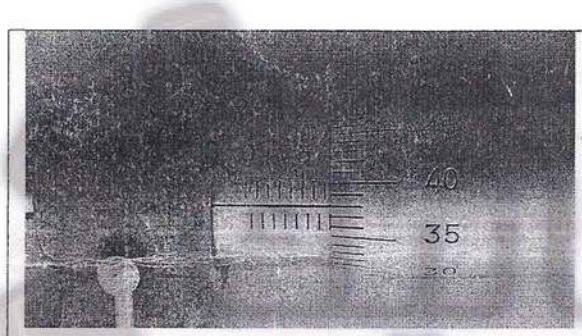
Physics 111
Final Exam
Time: 50 minutes;

الاسم بالعربية: _____ مدرس المختبر: د. اسحق راجي موعد المختبر : C - D

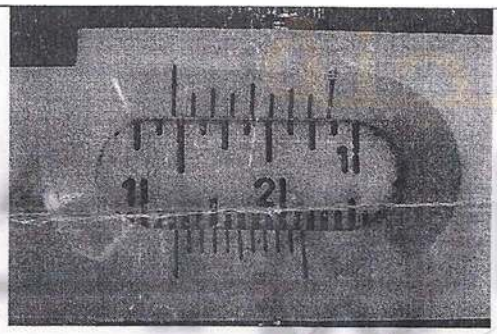
Q1: Express the results of the following calculations to the appropriate number of significant figures?

a) $0.42 \times 0.09 = 0,038$ (1) b) $\frac{1.36 \times 0.51}{2.6} = 0,27$ (1) c) $3.76 + 14.83 + 2.11 = 21$

Q2: What are the readings of the following instruments?



The reading = 5,88 ±0,005 mm (1)



The reading = 1,41 ±0,05 cm

Q3: State the two equations required to add perpendicular vectors?

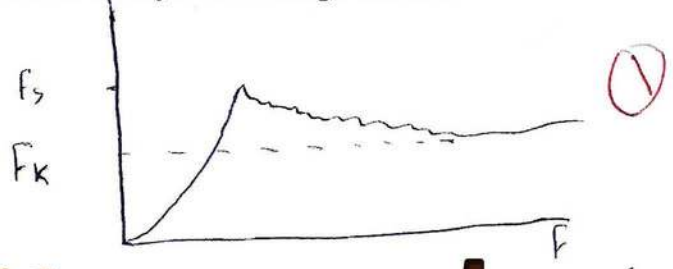
a) $a\sin 60\hat{i} + b\cos 60\hat{j}$ b) $a\sin 30\hat{i} + a\cos 30\hat{j}$

$= \sqrt{(x\hat{i})^2 + (y\hat{j})^2}$ (1)

Q4: State the purpose of the vectors experiment

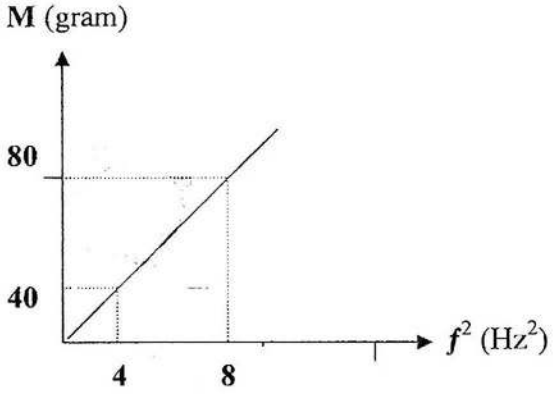
(1) equilibrium force and find R and add vector

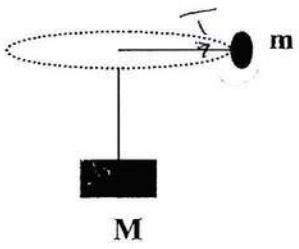
Q5: Sketch the curve demonstrating the dependence of the coefficients of friction on the force applied to move an object on a rough surface.



(1)

Q6: In the circular motion experiment, a student studied the relation between the hanging mass (**M**) and the square of the frequency (f^2) for rotational body (**m**) with the radius (**R=0.2 m**). A plotted data are shown in the figure bellow.





a) The mass (**m**) of the rotational body is:

$\frac{80-40}{8-4} = 10$

b) If **m=10 gram**, **M= 50 gram**, what is the frequency (in Hz) to make the body **m** rotate in a circular motion with the radius (**R=0.18 m**).

$f = \sqrt{\frac{mg}{4\pi^2 r}}$ $\sqrt{\frac{50 \times 9.8}{2 \times 3.14 \times 0.18}}$ $= 2,966$

Q7: Derive the equation determining the incline-angle an object placed on an adjustable rough incline begins to slide.